1. Install nginx and run nginx on port number 81.

```
       _  #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
ec2-user@ip-172-31-47-48 ~]$ sudo su -
root@ip-172-31-47-48 ~]# yum install nginx
mazon Linux 2023 Kernel Livepatch repository                        [===                                                                  ] ---  B/s |   0  B     --:-- EAm
on Linux 2023 Kernel Livepatch repository                                                                                  728 kB/s |  89 kB     00:00
ependencies resolved.
================================================================================================================================================================
 Package                                  Architecture                  Version                                      Repository                              Size
================================================================================================================================================================
nstalling:
 nginx                                    x86_64                        1:1.30.5-1.amzn2023.0.1                      amazonlinux                             34 k
nstalling dependencies:
 generic-logos-httpd                      noarch                        18.0.0-12.amzn2023.0.3                       amazonlinux                             19 k
 gperftools-libs                          x86_64                        2.9.1-1.amzn2023.0.3                         amazonlinux                            308 k
 libunwind                                x86_64                        1.4.0-5.amzn2023.0.3                         amazonlinux                             66 k
 nginx-core                               x86_64                        1:1.30.5-1.amzn2023.0.1                      amazonlinux                            711 k
 nginx-filesystem                         noarch                        1:1.30.5-1.amzn2023.0.1                      amazonlinux                             10 k
 nginx-mimetypes                          noarch                        2.1.49-3.amzn2023.0.3                        amazonlinux                             21 k

ransaction Summary
================================================================================================================================================================
nstall  7 Packages

otal download size: 1.1 M
nstalled size: 3.8 M
```

```
Complete!
[root@ip-172-31-47-48 ~]# sudo systemctl start nginx
[root@ip-172-31-47-48 ~]# sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 09:12:41 UTC; 10s ago
    Process: 26721 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 26722 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 26723 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 26724 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 57ms
     CGroup: /system.slice/nginx.service
             ├─26724 "nginx: master process /usr/sbin/nginx"
             ├─26725 "nginx: worker process"
             └─26726 "nginx: worker process"

Oct 14 09:12:41 ip-172-31-47-48.ap-southeast-2.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Oct 14 09:12:41 ip-172-31-47-48.ap-southeast-2.compute.internal nginx[26722]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Oct 14 09:12:41 ip-172-31-47-48.ap-southeast-2.compute.internal nginx[26722]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Oct 14 09:12:41 ip-172-31-47-48.ap-southeast-2.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-47-48 ~]# netstat -na | grep 80
tcp        0      0 0.0.0.0:80              0.0.0.0:*               LISTEN
tcp        0      0 172.31.47.48:55660      169.254.169.254:80      TIME_WAIT
tcp6       0      0 :::80                   :::*                    LISTEN
udp6       0      0 fe80::4ff:e5ff:fef3:546 :::*
unix  2      [ ACC ]     STREAM     LISTENING     6880     /run/systemd/private
unix  3      [ ]         STREAM     CONNECTED     3380
unix  3      [ ]         STREAM     CONNECTED     2680     /run/systemd/journal/stdout
[root@ip-172-31-47-48 ~]# lsof -i tcp:80
COMMAND   PID  USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
nginx   26724  root    8u  IPv4  31494      0t0  TCP *:http (LISTEN)
```

```
    worker_connections 1024;
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
                      '$status $body_bytes_sent "$http_referer" '
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    # Load modular configuration files from the /etc/nginx/conf.d directory.
    # See http://nginx.org/en/docs/ngx_core_module.html#include
    # for more information.
    include /etc/nginx/conf.d/*.conf;

    server {
        listen      81;
        listen       [::]:81;

^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket   M-B Previous
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-6 Copy         ^B Where Was     M-F Next
```

```
[root@ip-172-31-47-48 ~]# cd /var/log/
[root@ip-172-31-47-48 log]# ls
README  audit  chrony                cloud-init.log   dnf.log      hawkey.log  lastlog  private  sssd      wtmp
amazon  btmp   cloud-init-output.log  dnf.librepo.log  dnf.rpm.log  journal     nginx    sa       tallylog
[root@ip-172-31-47-48 log]# cd nginx
[root@ip-172-31-47-48 nginx]# ls
access.log  error.log
[root@ip-172-31-47-48 nginx]# cat error.log
2026/10/14 09:12:41 [notice] 26723#26723: using the "epoll" event method
2026/10/14 09:12:41 [notice] 26723#26723: nginx/1.30.5
2026/10/14 09:12:41 [notice] 26723#26723: built by gcc 11.5.0 20240719 (Red Hat 11.5.0-5) (GCC)
2026/10/14 09:12:41 [notice] 26723#26723: OS: Linux 6.18.54-121.166.amzn2023.x86_64
2026/10/14 09:12:41 [notice] 26723#26723: getrlimit(RLIMIT_NOFILE): 65535:65535
2026/10/14 09:12:41 [notice] 26724#26724: start worker processes
2026/10/14 09:12:41 [notice] 26724#26724: start worker process 26725
2026/10/14 09:12:41 [notice] 26724#26724: start worker process 26726
[root@ip-172-31-47-48 nginx]# sudo nano /etc/nginx/sites-available/default
[root@ip-172-31-47-48 nginx]# ls /etc/nginx/
conf.d      fastcgi.conf          fastcgi_params          koi-utf  mime.types          nginx.conf          scgi_params          uwsgi_params          win-utf
default.d  fastcgi.conf.default  fastcgi_params.default  koi-win  mime.types.default  nginx.conf.default  scgi_params.default  uwsgi_params.default
[root@ip-172-31-47-48 nginx]# sudo nano /etc/nginx/nginx.conf
[root@ip-172-31-47-48 nginx]# sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
[root@ip-172-31-47-48 nginx]# sudo systemctl restart nginx
[root@ip-172-31-47-48 nginx]# lsof -i tcp:81
COMMAND   PID  USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
nginx   27420  root    8u  IPv4  35413      0t0  TCP *:81 (LISTEN)
nginx   27420  root    9u  IPv6  35414      0t0  TCP *:81 (LISTEN)
nginx   27421 nginx    8u  IPv4  35413      0t0  TCP *:81 (LISTEN)
nginx   27421 nginx    9u  IPv6  35414      0t0  TCP *:81 (LISTEN)
nginx   27422 nginx    8u  IPv4  35413      0t0  TCP *:81 (LISTEN)
nginx   27422 nginx    9u  IPv6  35414      0t0  TCP *:81 (LISTEN)
[root@ip-172-31-47-48 nginx]#
```



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

Install Nginx and Run Nginx on Port 81

1. I updated the Amazon Linux packages.
2. I installed Nginx on the server.
3. I started the Nginx service.
4. I checked the Nginx service status to make sure it was running.
5. I opened the Nginx configuration file.
6. I changed the default Nginx listening port from **80 to 81**.
7. I saved the Nginx configuration file.
8. I tested the Nginx configuration to make sure there were no errors.
9. I restarted Nginx to apply the new port configuration.
10. I verified that Nginx was listening on **port 81**.
11. I allowed **TCP port 81** in the AWS EC2 Security Group.

12. I accessed Nginx from the browser using the **EC2 public IP and port 81**.
13. I verified that the **Nginx welcome page** was displayed successfully.

2. Deploy a sample index.html file on nginx.

```
[root@ip-172-31-47-48 ~]# ls
[root@ip-172-31-47-48 ~]# cd /usr/share/nginx/html/
[root@ip-172-31-47-48 html]# ls
404.html  50x.html  icons  index.html  nginx-logo.png  poweredby.png
[root@ip-172-31-47-48 html]# cat index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

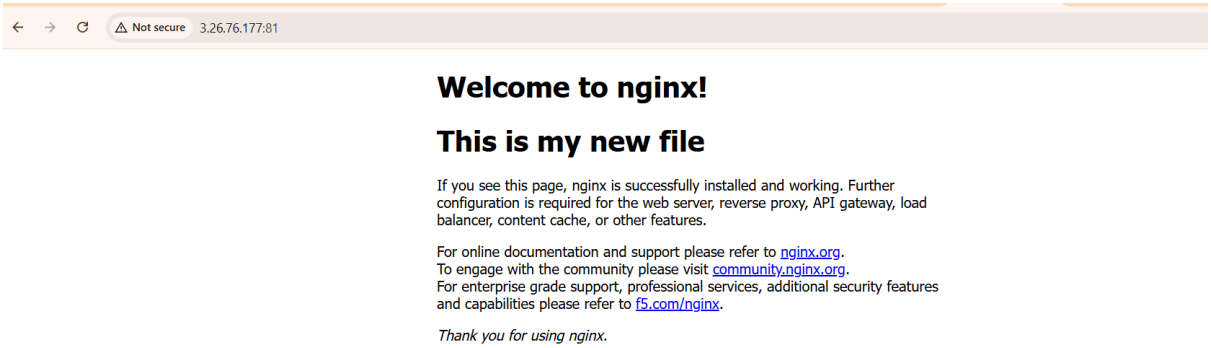
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[root@ip-172-31-47-48 html]# vi index.html
[root@ip-172-31-47-48 html]# sudo mv index.html index-old.html
```

```
[root@ip-172-31-47-48 html]# cat
^C
[root@ip-172-31-47-48 html]# cat index.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<h1>This is my new file</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[root@ip-172-31-47-48 html]# ls
404.html  50x.html  icons  index-old.html  index.html  nginx-logo.png  poweredby.png
```

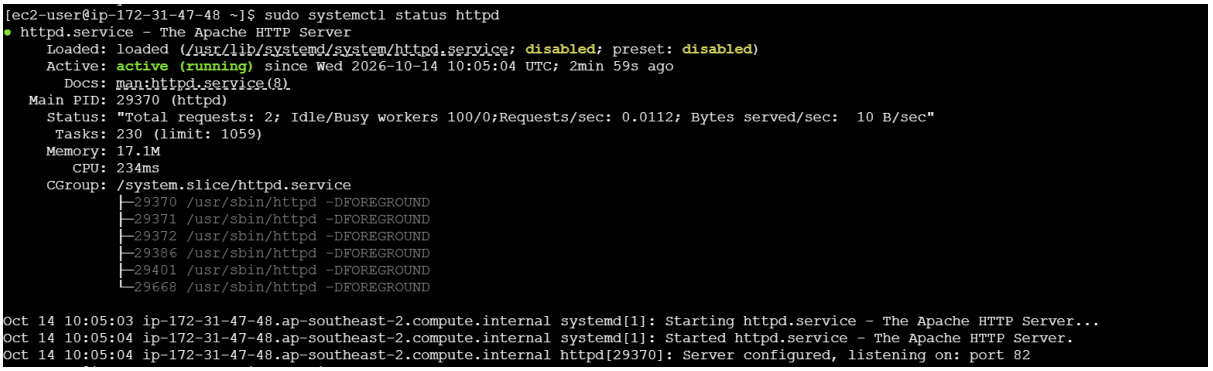


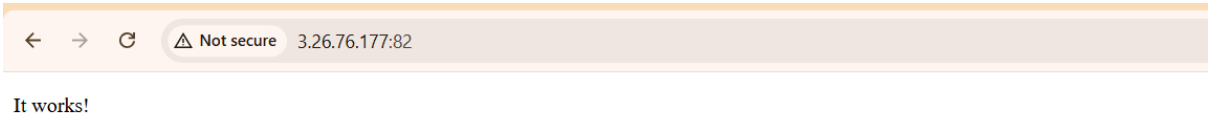
Deploy a Sample index.html File on Nginx

1. I navigated to the Nginx web root directory.
2. I created a new index.html file.

3. I added sample HTML content to the index.html file.
4. I saved the index.html file in the Nginx web directory.
5. I verified that the index.html file was created successfully.
6. I tested the webpage locally using the Nginx port.
7. I accessed the webpage using the EC2 public IP address and port 81.
8. I verified that my sample index.html page was displayed successfully in the browser.

3. Install Apache and run Apache on port number 82





It works!

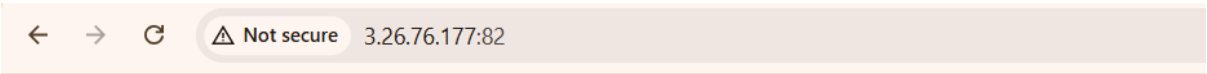
Install Apache and Run Apache on Port Number 82

1. I installed the Apache HTTP Server on the Linux server.
2. I opened the Apache configuration file.
3. I changed the default Apache port from 80 to 82.

4. I checked the Apache configuration for any syntax errors.
5. I restarted the Apache service to apply the port change.
6. I verified that Apache was running successfully on port 82.
7. I tested the Apache server locally using port 82.
8. I accessed Apache using the EC2 public IP address and port 82 in the browser.
9. I verified that the Apache default webpage was displayed successfully.

4. Deploy a sample index.html file on Apache

```
[ec2-user@ip-172-31-47-48 ~]$ cd /var/www/html
[ec2-user@ip-172-31-47-48 html]$ ls
[ec2-user@ip-172-31-47-48 html]$ sudo vi index.html
[ec2-user@ip-172-31-47-48 html]$ cat index.html
<!DOCTYPE html>
<html>
<head>
    <title>Amazon Linux Apache</title>
</head>
<body>
    <h1>Hello from Amazon Linux!</h1>
    <p>My sample index.html is deployed on Apache.</p>
</body>
</html>
[ec2-user@ip-172-31-47-48 html]$ ls -l /var/www/html/
total 4
-rw-r--r--. 1 root root 193 Oct 14 10:12 index.html
[ec2-user@ip-172-31-47-48 html]$ curl http://localhost:82
<!DOCTYPE html>
<html>
<head>
    <title>Amazon Linux Apache</title>
</head>
<body>
    <h1>Hello from Amazon Linux!</h1>
    <p>My sample index.html is deployed on Apache.</p>
</body>
</html>
[ec2-user@ip-172-31-47-48 html]$
```



Hello from Amazon Linux!

My sample index.html is deployed on Apache.

1. I navigated to the Apache web root directory.
2. I created a new index.html file.

3. I added sample HTML content to the index.html file.
4. I saved the index.html file in the Apache web directory.
5. I verified that the index.html file was created successfully.
6. I tested the webpage locally using Apache on **port 82**.
7. I accessed the webpage using the **EC2 public IP address and port 82**.
8. I verified that my sample index.html page was displayed successfully in the browser.

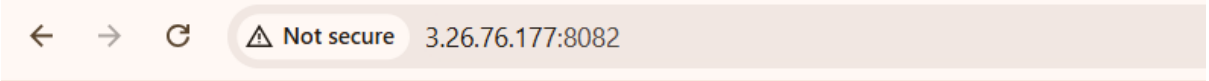
5. Install Apache tomcat on port number 8082

```
[root@ip-172-31-47-48 ~]# yum install tomcat10
Last metadata expiration check: 1:04:50 ago on Wed Oct 14 09:12:15 2026.
Dependencies resolved.
================================================================================
 Package                    Architecture  Version                    Repository     Size
================================================================================
Installing:
 tomcat10                   noarch        1:10.1.60-1.amzn2023.0.1   amazonlinux    94 k
Installing dependencies:
 cairo                      x86_64        1.18.0-4.amzn2023.0.3      amazonlinux   717 k
 ecj                        noarch        1:4.23-1.amzn2023.0.4      amazonlinux   2.4 M
 fonts-filesystem           noarch        1:2.0.5-12.amzn2023.0.2    amazonlinux   9.5 k
 google-noto-fonts-common   noarch        20240401-1.amzn2023.0.2    amazonlinux    17 k
 google-noto-sans-vf-fonts  noarch        20240401-1.amzn2023.0.2    amazonlinux   593 k
 graphite2                  x86_64        1.3.14-7.amzn2023.0.3      amazonlinux    97 k
 harfbuzz                   x86_64        7.0.0-2.amzn2023.0.2       amazonlinux   873 k
 java-17-amazon-corretto-headless x86_64  1:17.0.20+12-1.amzn2023.1  amazonlinux    91 M
 javapackages-filesystem    noarch        6.0.0-7.amzn2023.0.6       amazonlinux    12 k
 javapackages-tools         noarch        6.0.0-7.amzn2023.0.6       amazonlinux    25 k
 langpacks-core-font-en     noarch        3.0-21.amzn2023.0.4        amazonlinux    10 k
 libX11                     x86_64        1.8.10-2.amzn2023.0.2      amazonlinux   655 k
 libX11-common              noarch        1.8.10-2.amzn2023.0.2      amazonlinux   144 k
 libXau                     x86_64        1.0.11-6.amzn2023.0.1      amazonlinux    33 k
 libXext                    x86_64        1.3.6-1.amzn2023.0.1       amazonlinux    42 k
 libXrender                 x86_64        0.9.11-6.amzn2023.0.2      amazonlinux    28 k
 libjpeg-turbo              x86_64        2.1.4-2.amzn2023.0.5       amazonlinux   190 k
 libpng                     x86_64        2:1.6.37-10.amzn2023.0.13  amazonlinux   128 k
 libxcb                     x86_64        1.17.0-1.amzn2023.0.1      amazonlinux   235 k
 pixman                     x86_64        0.43.4-1.amzn2023.0.4      amazonlinux   296 k
 tomcat10-el-5.0-api        noarch        1:10.1.60-1.amzn2023.0.1   amazonlinux   109 k
 tomcat10-jsp-3.1-api       noarch        1:10.1.60-1.amzn2023.0.1   amazonlinux    77 k
 tomcat10-lib               noarch        1:10.1.60-1.amzn2023.0.1   amazonlinux   7.1 M
 tomcat10-servlet-6.0-api   noarch        1:10.1.60-1.amzn2023.0.1   amazonlinux   361 k
 xml-common                 noarch        0.6.3-56.amzn2023.0.2      amazonlinux    32 k
```

```
[root@ip-172-31-47-48 ~]# sudo systemctl start tomcat10
[root@ip-172-31-47-48 ~]# sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 10:17:51 UTC; 14s ago
   Main PID: 31302 (java)
      Tasks: 30 (limit: 1059)
     Memory: 59.3M
        CPU: 2.221s
     CGroup: /system.slice/tomcat10.service
             └─31302 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.Factory=org.apache.commons.dbcp.BasicDataSourceFactory --add-opens=java.base/java.lang=ALL-UNNAMED>

Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.067 INFO [main] org.apache.catalina.startup.VersionLoggerListener.lo>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.067 INFO [main] org.apache.catalina.startup.VersionLoggerListener.lo>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.073 INFO [main] org.apache.catalina.core.AprLifecycleListener.lifecy>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.079 INFO [main] org.apache.catalina.core.AprLifecycleListener.initia>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.560 INFO [main] org.apache.coyote.AbstractProtocol.init Initializing>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.622 INFO [main] org.apache.catalina.startup.Catalina.load Server ini>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.708 INFO [main] org.apache.catalina.core.StandardService.startIntern>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.712 INFO [main] org.apache.catalina.core.StandardEngine.startInterna>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.730 INFO [main] org.apache.coyote.AbstractProtocol.start Starting Pr>
Oct 14 10:17:52 ip-172-31-47-48.ap-southeast-2.compute.internal server[31302]: 14-Oct-2026 10:17:52.756 INFO [main] org.apache.catalina.startup.Catalina.start Server st>
lines 1-20/20 (END)
```

```
[ec2-user@ip-172-31-47-48 ~]$ ls -l /var/lib/tomcat10/webapps/
total 0
[ec2-user@ip-172-31-47-48 ~]$ sudo find / -type d -name webapps 2>/dev/null
/var/lib/tomcat10/webapps
[ec2-user@ip-172-31-47-48 ~]$ ls -l /var/lib/tomcat10/webapps
total 0
[ec2-user@ip-172-31-47-48 ~]$ ls -l /usr/share/tomcat10/webapps/ROOT/
ls: cannot access '/usr/share/tomcat10/webapps/ROOT/': No such file or directory
[ec2-user@ip-172-31-47-48 ~]$ sudo mkdir -p /usr/share/tomcat10/webapps/ROOT
[ec2-user@ip-172-31-47-48 ~]$ sudo vi /usr/share/tomcat10/webapps/ROOT/index.html
[ec2-user@ip-172-31-47-48 ~]$ sudo systemctl restart tomcat10
[ec2-user@ip-172-31-47-48 ~]$ curl http://localhost:8082/
<!DOCTYPE html>
<html>
<head>
    <title>My Tomcat Server</title>
</head>
<body>
    <h1>Welcome to Apache Tomcat</h1>
    <p>Tomcat is running successfully on port 8082.</p>
</body>
</html>
[ec2-user@ip-172-31-47-48 ~]$
```



Welcome to Apache Tomcat

Tomcat is running successfully on port 8082.

I installed Apache Tomcat 10 on the Linux server.
I opened the Tomcat configuration file.
I changed the default Tomcat port from 8080 to 8082.
I saved the Tomcat configuration after changing the port.
I started the Tomcat service.
I checked the Tomcat service status to make sure it was running.
I verified that Tomcat was listening on port 8082.
I tested the Tomcat server locally using port 8082.
I accessed Tomcat using the EC2 public IP address and port 8082 in the browser.
I verified that Tomcat was running successfully on port 8082.

6. Deploy a sample app on webapps

```
[ec2-user@ip-172-31-47-48 ~]$ cd /usr/share/tomcat10/webapps
[ec2-user@ip-172-31-47-48 webapps]$ ls
ROOT
[ec2-user@ip-172-31-47-48 webapps]$ sudo mkdir sampleapp
[ec2-user@ip-172-31-47-48 webapps]$ cd sampleapp
[ec2-user@ip-172-31-47-48 sampleapp]$ sudo vi index.html
[ec2-user@ip-172-31-47-48 sampleapp]$ sudo systemctl restart tomcat10
[ec2-user@ip-172-31-47-48 sampleapp]$ sudo systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 10:45:55 UTC; 10s ago
   Main PID: 34598 (java)
      Tasks: 30 (limit: 1059)
     Memory: 64.2M
        CPU: 4.344s
     CGroup: /system.slice/tomcat10.service
             └─34598 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.Factory=org.apache.commons.dbcp.BasicDataSourceFactory --add-opens=java.base/java.lang=ALL-UNNAMED>

Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.238 INFO [main] org.apache.catalina.core.StandardService.startIntern>
Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.241 INFO [main] org.apache.catalina.core.StandardEngine.startInterna>
Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.254 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto>
Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.991 INFO [main] org.apache.jasper.servlet.TldScanner.scanJars At lea>
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.068 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto>
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.070 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto>
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.359 INFO [main] org.apache.jasper.servlet.TldScanner.scanJars At lea>
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.362 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto>
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.367 INFO [main] org.apache.coyote.AbstractProtocol.start Starting Pr>
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.388 INFO [main] org.apache.catalina.startup.Catalina.start Server st>
[ec2-user@ip-172-31-47-48 sampleapp]$ curl http://localhost:8082/sampleapp/
<!DOCTYPE html>
<html>
<head>
    <title>Sample Tomcat App</title>
</head>
```

```
Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.238 INFO [main] org.apache.catalina.core.StandardService.startIntern
Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.241 INFO [main] org.apache.catalina.core.StandardEngine.startInterna
Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.254 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto
Oct 14 10:45:56 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:56.991 INFO [main] org.apache.jasper.servlet.TldScanner.scanJars At lea
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.068 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.070 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.359 INFO [main] org.apache.jasper.servlet.TldScanner.scanJars At lea
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.362 INFO [main] org.apache.catalina.startup.HostConfig.deployDirecto
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.367 INFO [main] org.apache.coyote.AbstractProtocol.start Starting Pr
Oct 14 10:45:57 ip-172-31-47-48.ap-southeast-2.compute.internal server[34598]: 14-Oct-2026 10:45:57.388 INFO [main] org.apache.catalina.startup.Catalina.start Server st
[ec2-user@ip-172-31-47-48 sampleapp]$ curl http://localhost:8082/sampleapp/
<!DOCTYPE html>
<html>
<head>
    <title>Sample Tomcat App</title>
</head>
<body>
    <h1>Welcome to My Sample Application</h1>
    <p>This application is deployed on Apache Tomcat.</p>
</body>
</html>
[ec2-user@ip-172-31-47-48 sampleapp]$
```

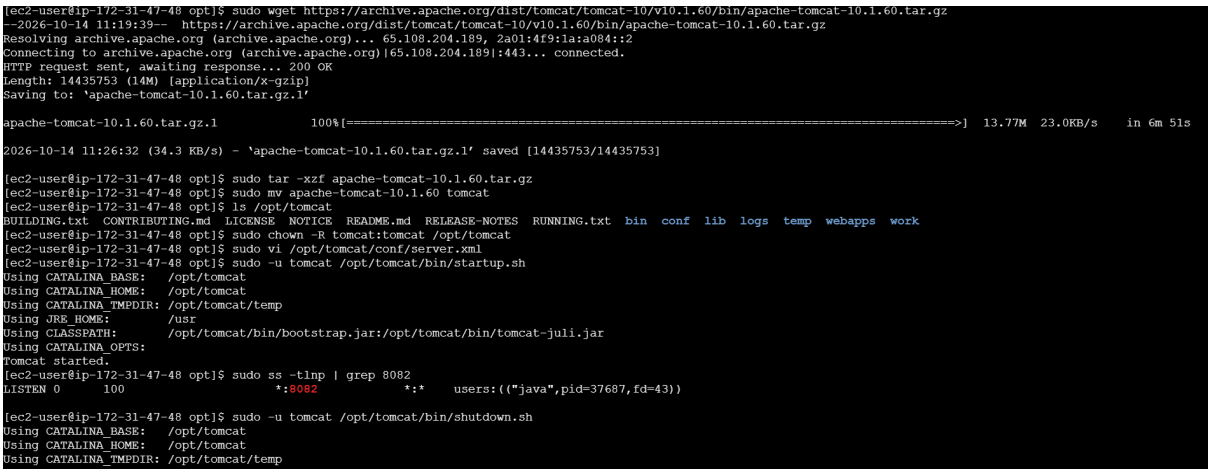


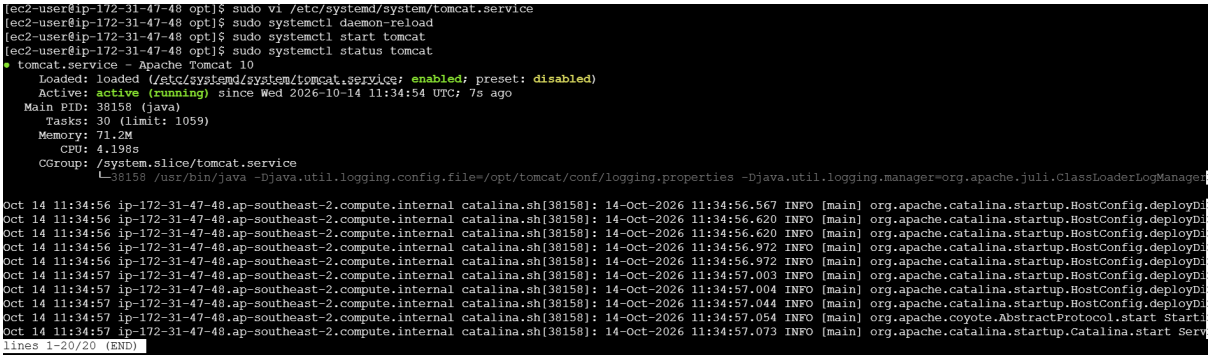
Welcome to My Sample Application

This application is deployed on Apache Tomcat.

I navigated to the Tomcat webapps directory.
I created a new folder for my sample application.
I created a sample index.html file inside the application folder.
I added sample content to the application.
I restarted the Tomcat service.
I tested the sample application using port 8082.
I accessed the application using the EC2 public IP address and port 8082.
I verified that my sample application was displayed successfully.

7. Create a tomcat.service file for tomcat





I created a tomcat.service file in the systemd directory.
I added the Tomcat service configuration to the file.
I configured the Tomcat user, group, Java environment, and Tomcat path.
I configured the service to start and stop Tomcat properly.
I saved the tomcat.service file.
I reloaded the systemd configuration.
I enabled the Tomcat service to start automatically after reboot.
I started the Tomcat service.
I checked the service status to verify that Tomcat was running successfully.

8. Configure HA Proxy server

```
  Running scriptlet: httpd-2.4.69-1.amzn2023.0.2.x86_64                                    13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                        1/13
  Verifying        : apr-util-1.6.5-1.amzn2023.0.1.x86_64                                   2/13
  Verifying        : apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64                              3/13
  Verifying        : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64                           4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                      5/13
  Verifying        : httpd-2.4.69-1.amzn2023.0.2.x86_64                                     6/13
  Verifying        : httpd-core-2.4.69-1.amzn2023.0.2.x86_64                                7/13
  Verifying        : httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch                          8/13
  Verifying        : httpd-tools-2.4.69-1.amzn2023.0.2.x86_64                               9/13
  Verifying        : libbrotli-1.1.0-1.amzn2023.0.1.x86_64                                 10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                  11/13
  Verifying        : mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64                              12/13
  Verifying        : mod_lua-2.4.69-1.amzn2023.0.2.x86_64                                  13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                     apr-util-1.6.5-1.amzn2023.0.1.x86_64
  apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64           apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch   httpd-2.4.69-1.amzn2023.0.2.x86_64
  httpd-core-2.4.69-1.amzn2023.0.2.x86_64             httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch
  httpd-tools-2.4.69-1.amzn2023.0.2.x86_64            libbrotli-1.1.0-1.amzn2023.0.1.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64
  mod_lua-2.4.69-1.amzn2023.0.2.x86_64

Complete!
[root@ip-172-31-33-18 ~]# vi /etc/hosts
[root@ip-172-31-33-18 ~]# ping load_balancer -c4
ping: load_balancer: Name or service not known
[root@ip-172-31-33-18 ~]# ping load_balancer -c 4
ping: load_balancer: Name or service not known
[root@ip-172-31-33-18 ~]# ping haproxy-server -c 4
PING haproxy-server (3.107.9.104) 56(84) bytes of data.

--- haproxy-server ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3112ms

[root@ip-172-31-33-18 ~]# systemctl start httpd
[root@ip-172-31-33-18 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 12:09:39 UTC; 12s ago
       Docs: man:httpd.service(8)
   Main PID: 26452 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 14.2M
        CPU: 76ms
     CGroup: /system.slice/httpd.service
             ├─26452 /usr/sbin/httpd -DFOREGROUND
             ├─26453 /usr/sbin/httpd -DFOREGROUND
             ├─26454 /usr/sbin/httpd -DFOREGROUND
             ├─26482 /usr/sbin/httpd -DFOREGROUND
             └─26499 /usr/sbin/httpd -DFOREGROUND

Oct 14 12:09:39 ip-172-31-33-18.ap-southeast-2.compute.internal systemd[1]: Starting httpd.service - T>
Oct 14 12:09:39 ip-172-31-33-18.ap-southeast-2.compute.internal systemd[1]: Started httpd.service - Th>
Oct 14 12:09:39 ip-172-31-33-18.ap-southeast-2.compute.internal httpd[26452]: Server configured, liste>
[root@ip-172-31-33-18 ~]#
```

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~
$ AMAyesha@LAPTOP-7RE57EC3 MINGW64 ~
bash: Ayesha@LAPTOP-7RE57EC3: command not found

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~
$ $
bash: $: command not found

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~
$ ssh -i C:\Users\Ayesha\Downloads\linux-testing.pem ec2-user@13.55.190.11
Warning: Identity file C:UsersAyeshaDownloadslinux-testing.pem not accessible: No such file or director
y.
The authenticity of host '13.55.190.11 (13.55.190.11)' can't be established.
ED25519 key fingerprint is SHA256:WdXOwXaScoRekL6GM/O9kZk/drfqxsRyXK+mgzZkwE8.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '13.55.190.11' (ED25519) to the list of known hosts.
ec2-user@13.55.190.11: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~
$ ls /c/Users/Ayesha/Downloads/linux-testing.pem
/c/Users/Ayesha/Downloads/linux-testing.pem

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~
$ chmod 400 /c/Users/Ayesha/Downloads/linux-testing.pem

Ayesha@LAPTOP-7RE57EC3 MINGW64 ~
$ ssh -i /c/Users/Ayesha/Downloads/linux-testing.pem ec2-user@13.55.190.11
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-33-18 ~]$ sudo -i
[root@ip-172-31-33-18 ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                   821 kB/s |  89 kB     00:00
Last metadata expiration check: 0:00:01 ago on Wed Oct 14 12:00:34 2026.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-33-18 ~]# yum install htppd -y
Last metadata expiration check: 0:00:40 ago on Wed Oct 14 12:00:34 2026.
No match for argument: htppd
Error: Unable to find a match: htppd
[root@ip-172-31-33-18 ~]# yum install httpd -y
Last metadata expiration check: 0:00:52 ago on Wed Oct 14 12:00:34 2026.
Dependencies resolved.
================================================================================
 Package                Architecture   Version                    Repository           Size
================================================================================
```

```
Ayesha@LAPTOP-7RE57EC3 MINGW64 ~
$ ssh -i /c/Users/Ayesha/Downloads/linux-testing.pem ec2-user@3.104.55.114
       #_
    ~\_  ####_        Amazon Linux 2023
   ~~  \_#####\
   ~~     \###|
   ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
    ~~       V~' '->
     ~~~         /
       ~~._.   _/
          _/ _/
        _/m/'
[ec2-user@ip-172-31-44-107 ~]$ sudo -i
[root@ip-172-31-44-107 ~]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                   792 kB/s |  89 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-44-107 ~]# yum install nginx -y
Last metadata expiration check: 0:02:21 ago on Wed Oct 14 12:14:18 2026.
Dependencies resolved.
================================================================================
 Package                 Architecture   Version                 Repository   Size
================================================================================
Installing:
 nginx                   x86_64         1:1.30.5-1.amzn2023.0.1 amazonlinux  34 k
Installing dependencies:
 generic-logos-httpd     noarch         18.0.0-12.amzn2023.0.3  amazonlinux  19 k
 gperftools-libs         x86_64         2.9.1-1.amzn2023.0.3    amazonlinux 308 k
 libunwind               x86_64         1.4.0-5.amzn2023.0.3    amazonlinux  66 k
 nginx-core              x86_64         1:1.30.5-1.amzn2023.0.1 amazonlinux 711 k
 nginx-filesystem        noarch         1:1.30.5-1.amzn2023.0.1 amazonlinux  10 k
 nginx-mimetypes         noarch         2.1.49-3.amzn2023.0.3   amazonlinux  21 k

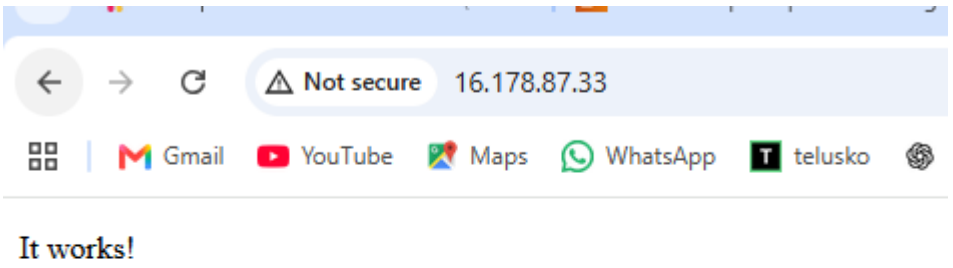
Transaction Summary
================================================================================
Install  7 Packages

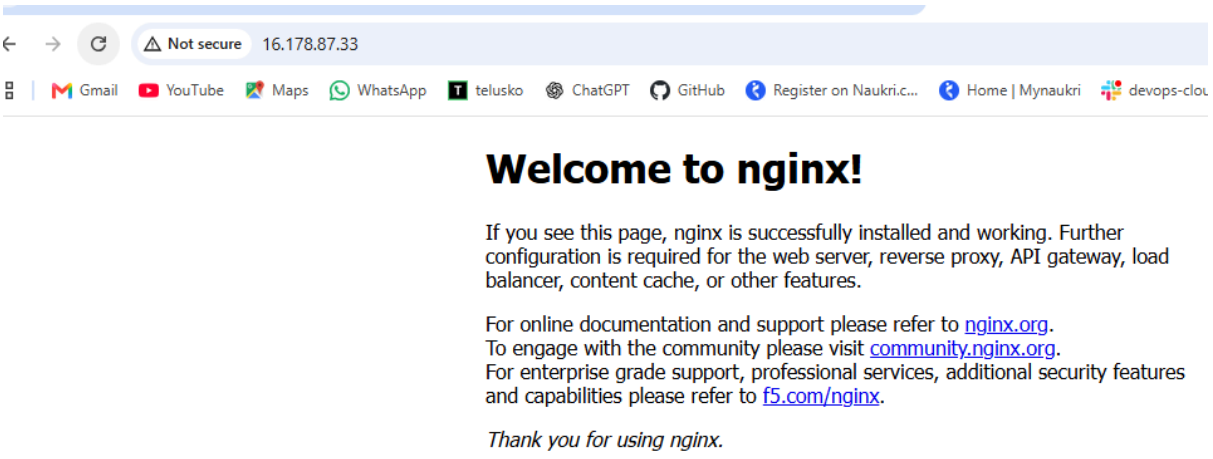
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm    703 kB/s |  19 kB     00:00
(2/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm                2.1 MB/s |  66 kB     00:00
(3/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm          8.2 MB/s | 308 kB     00:00
(4/7): nginx-1.30.5-1.amzn2023.0.1.x86_64.rpm                   1.1 MB/s |  34 kB     00:00
(5/7): nginx-filesystem-1.30.5-1.amzn2023.0.1.noarch.rpm        397 kB/s |  10 kB     00:00
(6/7): nginx-core-1.30.5-1.amzn2023.0.1.x86_64.rpm               17 MB/s | 711 kB     00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm         733 kB/s |  21 kB     00:00
--------------------------------------------------------------------------------
Total                                                           7.8 MB/s | 1.1 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Running scriptlet: nginx-filesystem-1:1.30.5-1.amzn2023.0.1.noarch        1/7
```

```
[root@ip-172-31-44-107 ~]# vi /etc/hosts
[root@ip-172-31-44-107 ~]# ping harproxy-server
ping: harproxy-server: Name or service not known
[root@ip-172-31-44-107 ~]# ping haproxy-server
PING haproxy-server (3.107.9.104) 56(84) bytes of data.
^C
--- haproxy-server ping statistics ---
193 packets transmitted, 0 received, 100% packet loss, time 199708ms

[root@ip-172-31-44-107 ~]# systemctl start nginx
[root@ip-172-31-44-107 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Wed 2026-10-14 12:23:47 UTC; 13min ago
   Main PID: 26984 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.5M
        CPU: 59ms
     CGroup: /system.slice/nginx.service
             ├─26984 "nginx: master process /usr/sbin/nginx"
             ├─26985 "nginx: worker process"
             └─26986 "nginx: worker process"

Oct 14 12:23:47 ip-172-31-44-107.ap-southeast-2.compute.internal systemd[1]: Starting nginx.service - T>
Oct 14 12:23:47 ip-172-31-44-107.ap-southeast-2.compute.internal nginx[26982]: nginx: the configuratio>
Oct 14 12:23:47 ip-172-31-44-107.ap-southeast-2.compute.internal nginx[26982]: nginx: configuration fi>
Oct 14 12:23:47 ip-172-31-44-107.ap-southeast-2.compute.internal systemd[1]: Started nginx.service - T>
[root@ip-172-31-44-107 ~]#
```





I installed and configured the HAProxy server.
I opened the HAProxy configuration file.
I configured HAProxy to listen for client requests.
I added the backend Nginx servers to the HAProxy configuration.
I configured load balancing between the backend servers.
I checked the HAProxy configuration for errors.
I started the HAProxy service.
I enabled HAProxy to start automatically after a reboot.
I checked the HAProxy service status.
I tested the connection through the HAProxy server.
I verified that the requests were distributed successfully between the backend servers.